



PSYCH/COMPSCI/BMI 841: Computational Cognitive Science

Credits: 3

Canvas Course URL: <https://canvas.wisc.edu/courses/446937>

Course Designations and Attributes: Counts toward 50% graduate coursework requirement

Meeting Times: Th 12:30-3:00pm

Meeting Location: Psychology 138 (and Zoom as necessary)

Optional Extra Mathematical Section: TBD

Instructional Mode: In-person

Credit Hour Compliance: *Traditional Carnegie Definition.* Approximately three hours of classroom (or online lecture) with readings, assignments, and projects that require over six hours per week.

INSTRUCTORS AND TEACHING ASSISTANTS

Professor: Joseph Austerweil (austerweil@wisc.edu)

Associate Professor of Psychology and Computer Science (affiliate)

Office: Psychology 328

Student Hours: Tu 2-3pm (zoom) and by appointment

In-person possible by appointment

OFFICIAL COURSE DESCRIPTION

Studies the biological and computational basis of intelligence, by combining methods from cognitive science, artificial intelligence, machine learning, computational biology, and cognitive neuroscience. Requires ability to program.

Unofficial Course Description: How is human memory and knowledge like a search engine and the Internet? Is human object recognition like a deep network? What can artificial intelligence and machine learning tell us about the mind? How can studying the mind help solve problems in machine learning? This course explores parallels between human cognition and contemporary research in computer science and statistics, emphasizing common problems. There will be a heavy emphasis on probabilistic modeling. There will be an optional weekly section to help students go through necessary mathematical concepts and hands-on help with implementing models using modern programming languages.

In addition to the topics mentioned above, the course will also cover shape and object recognition via deep networks and probabilistic grammars, human and machine knowledge

representation/retrieval, language learning and processing, pedagogy and machine teaching, reinforcement learning, hierarchical Bayesian models, Markov chain Monte Carlo, and more.

Official Requirements: None

Unofficial Prerequisites/Requirements: Prior experience with computational modeling and computer programming is required. Prior familiarity with linear algebra and probability theory is highly recommended. Instructor permission.

LEARNING OUTCOMES

Course Learning Outcomes Students who successfully complete the class will be able to

1. **understand** state-of-the-art machine learning techniques
2. **create** a probabilistic model of a problem solved by the human mind (hopefully one relevant to your research interests)
3. **analyze** how model assumptions (explicit or implicit) enable learning
4. **think critically** about the role of computational modeling in explaining human behavior

GRADING

<u>Requirement</u>	<u>Percentage of final grade</u>
Discussion posts	5% (5x 1% each)
Assignments	16% (#1 and 2 are 4%, 3 is 6%, and 4 is 2%)
Quizzes	16.5%
In-class discussion	5%
In-class presentations	10% (2x 5% each)
Project proposal	10%
Final project presentation	7.5%
Final project paper	30%

Overall course grade cutoffs will be

A: 93-100% AB: 88-92% B: 83-87% BC: 78-82% C: 70-77% D: 65-69% F: < 65%

You are guaranteed to get at least the grade corresponding to the sum of your scores on all assignments. If Prof. Austerweil decides on curving the overall grades, it can only be in your favor (you will get the same letter grade or better). It will never change the overall rank of students.

Extra credit policy. There is no extra credit in the class. No exceptions. Any curving will only happen at the end of the class.

DISCUSSION SESSIONS

There is no official discussion session for the course. However, there may be one early in the semester to help those with weaker programming and statistics background.

REQUIRED TEXTBOOK, SOFTWARE & OTHER COURSE MATERIALS

There is no required textbook. Instead, we will read journal articles and book chapters, which are available as PDFs online on Canvas through e-reserves.

For those looking for good reference textbooks, I recommend the following:

Introduction to Bayesian probability theory and data analysis:

Kruschke, J. K. (2016). *Doing Bayesian data analysis: A tutorial with R, JAGS, and STAN*. Burlington, MA: Academic Press.

Probabilistic modeling:

Murphy, K. P. (2012). *Machine Learning: A Probabilistic Perspective*. Cambridge, MA: MIT Press.

Both probabilistic modeling and deep learning:

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. Cambridge, MA: MIT Press.

EXAMS, QUIZZES, PAPERS & OTHER MAJOR GRADED WORK

The final project for the class will be an independent research project presenting a simple experiment, testing a new model, or analyzing an existing model. Students will work with me individually to develop their own project. Halfway through the course, you will submit a 1 page (single-spaced) project proposal that explains what the critical question that the project is examining, a short background on previous work on the question, and the methods for examining the question (**due 03/12 at 8:59pm**). The project will culminate in an (at most) 6 page (single-spaced) paper (**due 05/06 at 8:59pm**) and a 10 minute class presentation (05/01). Proposal, presentation, and paper should be submitted through Canvas.

HOMEWORK & OTHER ASSIGNMENTS

Assignments. There are four programming assignments (submitted through Canvas). The goal of these is to solidify your knowledge of the computational topics discussed in classes by implementing them. Interactive jupyter notebooks are provided for Python and R through the Canvas portal. Although it is not certain, the professor is hoping to provide support for Julia this semester as well. If you are interested in using a different language, please let me know and we can discuss it.

Discussion posts. You are expected to submit a short (50-200 word) reaction to the reading for the week on the discussion board (Discussion posts). These are due by 8:59pm on Tuesday of each week. The first one is **due on 02/03** and the last one is **due on 04/21**. You are only required to do 5 of them (out of 11 total). They are pass/fail and submitted through Canvas.

Quizzes. Modules for each week have quizzes interwoven into them. These include multiple choice questions and short programming or mathematical exercises. The class transitions from using more quizzes to assignments to the final project as it progresses.

Class presentations. You will be expected to give three class presentations over the course of the semester. Two of the presentations will be to provide the background and lead a discussion on a reading for the class. In these presentations, students first present the major results of the

paper, and then on the last slide have 4-6 discussion questions (to aid in guiding the discussion of the paper). Having controversial and/or provocative questions is encouraged! Student presentations are intended to be in-person during class time. Some of these may be conducted via Zoom during class time or asynchronously when necessary. **Please note: The slides for all class presentations are expected to be uploaded to Canvas by 8:59pm the same day as the presentation.**

Students will sign up for papers during the first few classes. It is your responsibility to make sure you are signed up to present two papers. If you are auditing the course, you are required to present one paper. The third presentation will be to present your final project to the class.

Please look at the tips for class presentations pdf for guidelines in making your presentation.

EMAIL POLICY

I will do my best to respond to emails within 36 hours (48 hours if you send it late Friday or over the weekend). Please do your best to keep them clear and concise (in fact, this is in your best interest because we will probably respond faster).

LATE POLICIES

Discussion posts. Late discussion posts will not be accepted. You only need to do 5 of 11, so if you need to be late one week, please do a discussion post for a different week. Also, please do not wait until the very end of the course to complete these (you may get sick then and have trouble completing it).

Quizzes. Quizzes are administered online. You have 72 hours to take them from when they are released. As there is a lot of flexibility for how and when you take the quizzes, after the due date late quizzes excusal will only be accepted in very rare circumstances (e.g., a serious unexpected emergency, such as hospitalization). You must contact Prof. Joe at least 24 hours prior to quiz deadline for consideration.

Please check all of the dates of quizzes now for conflicts. By taking the course and not reaching out to the professor within the first two weeks, you confirm that you have no conflicts with the quiz date/time.

Assignments. You get three “free” late days for assignments throughout the semester. They can be distributed across the four assignments in any manner (using all three for one problem set or one day for three assignments are both fine). They are not prorated (e.g., if you turn in assignment 12 hours late that counts as one of your days). After the three free late days for the semester are used, any assignments will be deducted 20% per day late (again not prorated, so 12 hours late would be a 20% deduction).

Proposal and paper. No lateness will be accepted for these assignments. Each will have a 20% deduction for every 24 hours late (not prorated, so 12 hours late would be a 20%

deduction). If you have extenuating circumstances, please contact Professor Austerweil to discuss them.

PRIVACY OF STUDENT RECORDS and the USAGE of AUDIO RECORDED LECTURES

See information about [privacy of student records and the usage of audio-recorded lectures](#).

Usage of Audio Recorded Lectures Statement

Lecture materials and recordings for PSYC/BMI/COMPSCI 841 are protected intellectual property at UW-Madison. Students in this course may use the materials and recordings for their personal use related to participation in this class. Students may also take notes solely for their personal use. If a lecture is not already recorded, you are not authorized to record my lectures without my permission unless you are considered by the university to be a qualified student with a disability requiring accommodation. [Regent Policy Document 4-1] Students may not copy or have lecture materials and recordings outside of class, including posting on internet sites or selling to commercial entities. Students are also prohibited from providing or selling their personal notes to anyone else or being paid for taking notes by any person or commercial firm without the instructor's express written permission. Unauthorized use of these copyrighted lecture materials and recordings constitutes copyright infringement and may be addressed under the university's policies, UWS Chapters 14 and 17, governing student academic and non-academic misconduct.

In light of potential changes to University of Wisconsin-Madison's copyright policies due to ongoing negotiations with the Universities of Wisconsin Regents, all course materials are considered part of Professor Austerweil's scholarly work.

STUDENTS' RULES, [RIGHTS & RESPONSIBILITIES](#)

By enrolling in this course, each student assumes the responsibilities of an active participant in UW-Madison's community of scholars in which everyone's academic work and behavior are held to the highest academic integrity standards. Academic misconduct compromises the integrity of the university. Cheating, fabrication, plagiarism, unauthorized collaboration, and helping others commit these acts are examples of academic misconduct, which can result in disciplinary action. This includes but is not limited to failure on the assignment/course, disciplinary probation, or suspension. Substantial or repeated cases of misconduct will be forwarded to the Office of Student Conduct & Community Standards for additional review. For more information, refer to studentconduct.wiscweb.wisc.edu/academic-integrity/. **Please note that in the case of plagiarism, you will be held to a more stringent standard than the standard university policy.**

Plagiarism policy. All assignments will be turned in through Canvas and automatically checked for plagiarism. Plagiarism is strictly prohibited and will be dealt with in a case-by-case basis. **NOTE: For the purposes of this class, self-plagiarism is plagiarism and thus, prohibited.** However, the course should be useful for your research career. So, you can turn in a final paper that also gets submitted to a conference. But, you cannot turn in a paper you used for a previous course. Please check with Professor Austerweil if you're concerned or confused.

Generative AI Policy. Over the last few years, the ease of use and capabilities of modern AI technology has reached an unprecedented level of performance. Broadly, you are welcome to use artificial intelligence (AI) tools and applications (such as ChatGPT, DALL-E, etc.) in this class as they support the learning objectives of this course. For technical issues, they have become an indispensable tool and banning their use would be counterproductive from both pedagogical and research standpoints. However, **you are prohibited from asking quiz or assignment questions directly to them. Your answers to quizzes and assignments can be informed by AI tools, but the majority of an answer should be written in your own words. For other parts of the class, you are expected to use the following policy.** Your use of AI tools must be documented and cited to conform to this course's expectations. **Any documentation or citations included with your work will not be counted against any word/page limits. Please be aware you are responsible for the information you submit based on an AI query** (e.g., ensure your professor has allowed you to publicly post course content such as assignment or assessment prompts and that the AI generated results do not contain misinformation or unethical content).

NOTE: Given the rate of improvement over the last few years, it is hard to predict how AI tools will change over the semester. If a major disruption occurs, this policy is subject to change. Any changes will be clearly discussed and communicated through lecture and online announcements.

Ethics of being a student in the Department of Psychology

The members of the faculty of the Department of Psychology at UW-Madison uphold the highest ethical standards of teaching and research. They expect their students to uphold the same standards of ethical conduct. By registering for this course, you are implicitly agreeing to conduct yourself with the utmost integrity throughout the semester.

In the Department of Psychology, acts of academic misconduct are taken very seriously. Such acts diminish the educational experience for all involved – students who commit the acts, classmates who would never consider engaging in such behaviors, and instructors. Academic misconduct includes, but is not limited to, cheating on assignments and exams, stealing exams, sabotaging the work of classmates, submitting fraudulent data, plagiarizing the work of classmates or published and/or online sources, acquiring previously written papers and submitting them (altered or unaltered) for course assignments, collaborating with classmates when such collaboration is not authorized, and assisting fellow students in acts of misconduct. Students who have knowledge that classmates have engaged in academic misconduct should report this to the instructor.

ACADEMIC INTEGRITY

By enrolling in this course, each student assumes the responsibilities of an active participant in UW-Madison's community of scholars in which everyone's academic work and behavior are held to the highest academic integrity standards. Academic misconduct compromises the integrity of the university. Cheating, fabrication, plagiarism, unauthorized collaboration, and helping others commit these acts are examples of academic misconduct, which can result in

disciplinary action. This includes but is not limited to failure on the assignment/course, disciplinary probation, or suspension. Substantial or repeated cases of misconduct will be forwarded to the Office of Student Conduct & Community Standards for additional review. For more information, refer to <https://conduct.students.wisc.edu/academic-misconduct/>.

ACCOMMODATIONS POLICIES

McBurney Disability Resource Center syllabus statement: “The University of Wisconsin-Madison supports the right of all enrolled students to a full and equal educational opportunity. The Americans with Disabilities Act (ADA), Wisconsin State Statute (36.12), and UW-Madison policy (Faculty Document 1071) require that students with disabilities be reasonably accommodated in instruction and campus life. Reasonable accommodations for students with disabilities is a shared faculty and student responsibility. Students are expected to inform faculty [me] of their need for instructional accommodations by the end of the third week of the semester, or as soon as possible after a disability has been incurred or recognized. Faculty [I], will work either directly with the student [you] or in coordination with the McBurney Center to identify and provide reasonable instructional accommodations. Disability information, including instructional accommodations as part of a student's educational record, is confidential and protected under FERPA.” <http://mcburney.wisc.edu/facstaffother/faculty/syllabus.php>

UW-Madison students who have experienced sexual misconduct (which can include sexual harassment, sexual assault, dating violence and/or stalking) also have the right to request academic accommodations. This right is afforded them under Federal legislation (Title IX). Information about services and resources (including information about how to request accommodations) is available through Survivor Services, a part of University Health Services: <https://www.uhs.wisc.edu/survivor-services/> .

DIVERSITY & INCLUSION

Institutional statement on diversity: “Diversity is a source of strength, creativity, and innovation for UW-Madison. We value the contributions of each person and respect the profound ways their identity, culture, background, experience, status, abilities, and opinion enrich the university community. We commit ourselves to the pursuit of excellence in teaching, research, outreach, and diversity as inextricably linked goals.

The University of Wisconsin-Madison fulfills its public mission by creating a welcoming and inclusive community for people from every background – people who as students, faculty, and staff serve Wisconsin and the world.” <https://diversity.wisc.edu/>

Complaints

Occasionally, a student may have a complaint about a TA or course instructor. If that happens, you should feel free to discuss the matter directly with the TA or instructor. If the complaint is about the TA and you do not feel comfortable discussing it with the individual, you should discuss it with the course instructor. Complaints about mistakes in grading should be resolved with the TA and/or instructor in the great majority of cases. If the complaint is about the

instructor (other than ordinary grading questions) and you do not feel comfortable discussing it with the individual, make an appointment to speak to the Associate Chair for Graduate Studies, Professor Shawn Green, cshawn.green@wisc.edu.

If you have concerns about climate or bias in this class, or if you wish to report an incident of bias or hate that has occurred in class, you may contact the Chair of the Department, Professor Allyson Bennett (allyson.j.bennett@wisc.edu) or the Chair of the Psychology Department Climate & Diversity Committee, Martha Alibali (martha.alibali@wisc.edu). You may also use the University's bias incident reporting system, which you can reach at the following link: <https://osas.wisc.edu/report-an-issue/bias-or-hate-reporting/>.

CONCERNS ABOUT SEXUAL MISCONDUCT

All students deserve to be safe and respected at UW-Madison. Unfortunately, we know that sexual and relationship violence do happen here. Free, confidential resources are available on and off campus for students impacted by sexual assault, sexual harassment, dating violence, and stalking (regardless of when the violence occurred). You don't have to label your experience to seek help. Friends of survivors can reach out for support too. A list of resources can be found at <https://www.uhs.wisc.edu/survivor-resources/>

If you wish to speak to someone in the Department of Psychology about your concerns, you may contact the Chair of the Department, Professor Allyson Bennett (allyson.j.bennett@wisc.edu) or the Associate Chair of Graduate Studies, Professor Shawn Green, cshawn.green@wisc.edu).

Please note that all of these individuals are Responsible Employees (<https://compliance.wisc.edu/titleix/mandatory-reporting/#responsible-employees>).

Schedule of classes and readings:

Week 1: Course Introduction/Admin and Intro to Basic Bayes (01/23)

Rahwan, I., Cebrian, M., Obradovich, N., Bongard, J., Bonnefon, J. F., Breazeal, C., ... & Jennings, N. R. (2019). Machine behaviour. *Nature*, 568(7753), 477-486.

Anderson, E. (2020, July). Controversial Detroit facial recognition got him arrested for a crime he didn't commit. *Detroit Free Press*, 1-4.

Austerweil, J. L. (in prep). Introduction to Bayesian theory. *Unpublished Manuscript*. (read until Continuous probability).

Week 2: Basic Bayes Continued (01/30)

Jordan, M. (1988). An Introduction to Linear Algebra in Parallel Distributed Processing. In D. E. Rumelhart & J. L. McClelland (Eds.), *Parallel distributed processing* (Vol. 1, p. 365-422). Cambridge, MA: MIT Press.

Austerweil, J. L. (in prep). Introduction to Bayesian theory. *Unpublished Manuscript*. (read Continuous probability and expectations. Stop at Gaussian-Gaussian posterior update).

(optional: extra prep): Etz, A., & Vandekerckhove, J. (2018). Introduction to Bayesian Inference for Psychology. *Psychonomic Bulletin & Review*, 25, 5-34.

Week 3: Conjugate Bayesian Models and Topic Models (02/06)

Note: Prof. Joe is on a work trip to Japan this week. Students presenting this week will need to record their presentation for the class to view. Instead of in-person class on Thursday 02/06, we will use an asynchronous format for this week only.

No OH on Tuesday. Any OH this week will be by appointment and virtual.

Austerweil, J. L. (in prep). Introduction to Bayesian theory. *Unpublished Manuscript*. (read 14, 15, 18-20; skim 16, 17). *Optional*: Read 16-17 in depth. It provides a full derivation for the Gaussian-Gaussian posterior update.

Körding, K. P., & Wolpert, D. M. (2004). Bayesian integration in sensorimotor learning. *Nature*, 427, 244-247.

Griffiths, T. L., & Tenenbaum, J. B. (2001). Randomness and Coincidences: Reconciling Intuition and Probability Theory. *Proceedings of the Twenty-Third Annual Conference of the Cognitive Science Society*.

Griffiths, T. L., & Steyers, M. (2000). A probabilistic approach to semantic representation. In *Proceedings of the 24th Annual Meeting of the Cognitive Science Society*.

[Note: Skim the section on how Gibbs sampling is used to estimate the distributions]

Hornsby, A. N., Evans, T., Riefer, P. S., Prior, R., & Love, B. C. (2020). Conceptual Organization is Revealed by Consumer Activity Patterns. *Computational Brain & Behavior*, 3, 162-173.

First Possible Discussion Post Due 02/03 at 8:59pm

Week 4: Bayesian generalization and Hierarchical Bayesian modeling (02/13)

No OH on Tuesday (in flight back to US). OH by appointment this week, though in-person will be possible later in the week.

Tenenbaum, J. B. (1999). Bayesian modeling of human concept learning. In *Advances in Neural Information Processing Systems*, 11.

Tenenbaum, J. B., & Xu, F. (2000). Word learning as Bayesian inference. In *Proceedings of the 22nd Annual Conference of the Cognitive Science Society* (pp. 517-522).

Austerweil, J. L., & Griffiths, T. L. (2010). Learning hypothesis spaces and dimensions through concept learning. In *Proceedings of the 32nd Annual Meeting of the Cognitive Science Society* (pp. 73-78).

Navarro, D. J., Lee, M. D., Dry, M. J., & Schultz, B. G. (2008). Extending and testing the Bayesian theory of generalization. In *Proceedings of the 30th Annual Meeting of the Cognitive Science Society* (pp. 1746-1751).

Week 5: Hierarchical Bayesian Models Cont'd and Bayesian Networks (02/20)

Kemp, C., Perfors, A., & Tenenbaum, J. B. (2007). Learning overhypotheses with hierarchical Bayesian models. *Developmental science*, 10(3), 307-321.

[Note: Don't worry about the extended model that includes the Chinese restaurant process – We will cover that in Bayesian nonparametrics]

Griffiths, T. L., & Steyvers, M. (2003). Prediction and semantic association. *Advances in Information Processing Systems* 15.

[Note: Don't worry about sections 5 or 6 if they are too challenging]

Zhu, X., Rogers, T. T., Qian, R., & Kalish, C. W. (2007). Humans perform semi-supervised learning too. In *The 22nd AAAI Conference on Artificial Intelligence* (pp. 864-869). AAAI.

De Zoete, J., Fenton, N., Noguchi, T., & Lagnado, D. (2019). Resolving the so-called “probabilistic paradoxes in legal reasoning” with Bayesian networks. *Science & Justice*, 59, 367-379.

Assignment 1 (Clusters) due on 02/19 at 8:59pm.

Week 6: Bayesian Networks Cont'd and Causal Bayesian Networks (02/27)

Lagnado, D. A., & Sloman, S. A. (2002). Learning Causal Structure. In *Proceedings of the 24th Annual Conference of the Cognitive Science Society*.

Tenenbaum, J. B., & Griffiths, T. L. (2001). Structure learning in human causal induction. In *Advances in Neural Information Processing Systems* (pp. 59-65).

Griffiths, T. L., Baraff, E. R., & Tenenbaum, J. B. (2004). Using physical theories to infer hidden causal structure. *Proceedings of the 26th Annual Conference of the Cognitive Science Society*.

[Note: The Nitro X section involves events occurring over time. If you are unfamiliar with Poisson Processes, feel free to skim that portion.]

Gerstenberg, T., Goodman, N. D., Lagnado, D. A., & Tenenbaum, J. B. (2015). How, whether, why: Causal judgments as counterfactual contrasts. In D. C. Noelle, R. Dale, A. S. Warlaumont, J. Yoshimi, T. Matlock, C. D. Jennings, & P. P. Maglio (Eds.), *Proceedings of the 37th annual meeting of the cognitive science society* (pp. 1919-1924). Austin, TX: Cognitive Science Society.

Week 7: Markov Chains and Network Theory (03/06)

Mukherjee, S., Romero, D. M., Jones, B., & Uzzi, B. (2017). The nearly universal link between the age of past knowledge and tomorrow's breakthroughs in science and technology: The hotspot. *Science Advances*, 3, e1601315.

Griffiths, T. L., Steyvers, M., & Firl, A. (2007). Google and the mind: Predicting fluency with PageRank. *Psychological Science*, 18(12), 1069-1076.

Abbott, J. T., Austerweil, J. L., & Griffiths, T. L. (2012). Human memory search as a random walk in a semantic network. *Advances in Neural Information Processing Systems*, 25 (pp. 3041-3049).

Zemla, J. C., & Austerweil, J. L. (2019). Analyzing knowledge retrieval impairments with Alzheimer's disease using network analyses. *Complexity*. doi: 10.1155/2019/4203158

Assignment 2 (Generalization) due on 03/05 at 8:59pm.

Week 8: Approximation: Monte Carlo (03/13):

Shi, L., Feldman, N. H., & Griffiths, T. L. (2008). Performing Bayesian inference with exemplar models. In V. Sloutsky, B. Love, and K. McRae (Eds.), *Proceedings of the 30th Annual Conference of the Cognitive Science Society* (pp. 745-750).

Vul, E., Goodman, N. D., Griffiths, T. L., & Tenenbaum, J. B. (2009). One and done? Optimal decisions from very few samples. In *Proceedings of the 31st Annual Conference of the Cognitive Science Society* (pp. 66-72).

Lieder, F., Hsu, M., & Griffiths, T. L. (2014). The high availability of extreme events serves resource-rational decision-making. In P. Bello, M. Guarini, M. McShane, & B. Scassellati (Eds.), *Proceedings of the 36th Annual Meeting of the Cognitive Science Society* (pp. 2567-2572).

Sanborn, A. N. & Griffiths, T. L. (2007). Markov chain Monte Carlo with people. In *Advances in Neural Information Processing Systems* (pp. 1265-1272).

Project proposal due 03/12 at 8:59pm

Week 9: Statistical Decision Theory, Markov Decision Processes, and Reinforcement Learning (03/20):

Schultz, W., Dayan, P., & Montague, P. R. (1997). A Neural Substrate of Prediction and Reward. *Science*, 275, 1593-1599.

Ho, M. K., Littman, M. L., Cushman, F., & Austerweil, J. L. (2015). Teaching with rewards and punishments: Reinforcement or communication?. In *Proceedings of the 37th Annual Meeting of the Cognitive Science Society*. Pasadena, CA.

Daw, N. D., Niv, Y., & Dayan, P. (2005). Uncertainty-based competition between prefrontal and dorsolateral striatal systems for behavioral control. *Nature Neuroscience* 8(12), 1704-1711.

Ho, M. K., Littman, M. L., & Austerweil, J. L. (2017). Teaching by intervention: Working backwards, undoing mistakes, or correcting mistakes? In G. Gunzelmann, A. Howes, T. Tenbrink, & E. Davelaar (Eds.), *Proceedings of the 39th Annual Meeting of the Cognitive Science Society* (pp. 526-531). Austin, TX: Cognitive Science Society.

No class 03/27 – Spring Break

Week 10: (Inverse) Reinforcement Learning Cont'd (04/03)

Baker, C. L., Tenenbaum, J. B., & Saxe, R. R. (2007). Goal Inference as Inverse Planning. In D. S. McNamara & J. G. Trafton (Eds.), *Proceedings of the 29th Annual Cognitive Science Society* (pp. 779-784). Austin, TX: Cognitive Science Society.

Ho, M. K., Littman, M. L., MacGlashan, J., Cushman, F., & Austerweil, J. L. (2016). Showing versus doing: Teaching by demonstration. In D. D. Lee, M. Sugiyama, U. V. Luxburg, I. Guyon, & R. Garnett (Eds.), *Advances in Neural Information Processing Systems* (pp. 3027-3035). Red Hook, NY: Curran Associates, Inc.

Zühlsdorff, K., Verdejo-Román, J., Clark, L., Albein-Urios, N., Soriano-Mas, C., Cardinal, R. N., ... & Kanen, J. W. (2024). Computational modelling of reinforcement learning and functional neuroimaging of probabilistic reversal for dissociating compulsive behaviours in gambling and cocaine use disorders. *BJPsych Open*, 10(1), e8.

Mujica-Parodi, L. R., & Strey, H. H. (2020). Making sense of computational psychiatry. *International Journal of Neuropsychopharmacology*, 23(5), 339-347.

Week 11: Bayesian Nonparametrics (04/10)

Introduction (no discussion posts): Austerweil, J. L., Gershman, S. J., Tenenbaum, J. B., & Griffiths, T. L. (2015). Structure and flexibility in Bayesian models of cognition. In J. R. Busemeyer, Z. Wang, J. T. Townsend, & A. Eidels (Eds.), *Oxford Handbook of Computational and Mathematical Psychology* (pp. 187-208). Oxford University Press.

Gershman, S. J., Blei, D. M., & Niv, Y. (2010). Context, learning, and extinction. *Psychological Review*, 117(1), 197-209.

Kemp, C., & Tenenbaum, J. B. (2003). Theory-based Induction. In *proceedings of the Twenty-Fifth Annual Meeting of the Cognitive Science Society*.

Austerweil, J. L., & Griffiths, T. L. (2008). Analyzing human feature learning as nonparametric Bayesian inference. In D. Koller, D. Schuurmans, Y Bengio, & L. Buttou (Eds.) *Advances in Neural Information Processing Systems* (pp. 97-104).

Griffiths, T. L., Lucas, C., Williams, J., & Kalish, M. (2008). Human function learning with Gaussian processes. In D. Koller, D. Schuurmans, Y Bengio, & L. Buttou (Eds.) *Advances in Neural Information Processing Systems*.

Assignment 3 (Monte Carlo Approximation) due on 04/09 at 8:59pm

Week 12: Convolutional and Recurrent Deep Neural Networks (04/17)

Lake, B. M., Zaremba, W., Fergus, R., & Gureckis, T. M. (2015). Deep Neural Networks Predict Category Typicality Ratings for Images. In R. Dale, C. Jennings, P. Maglio, T. Matlock, D. Noelle, & A. Warlaumont (Eds) *Proceedings of the 37th Annual Meeting of the Cognitive Science Society* (pp. 1243-1248).

Ritter, S., Barrett, D. G. T., Santoro, A., & Botvinick, M. M. (2017). Cognitive Psychology for Deep Neural Networks: A Shape Bias Case Study. In D. Precup & Y. W. Teh (Eds.), *Proceedings of the 34th International Conference on Machine Learning* (pp. 2940-2949).

Linzen, T., & Leonard, B. (2018). Distinct patterns of syntactic agreement errors in recurrent networks and humans. In *Proceedings of the 40th Annual Meeting of the Cognitive Science Society* (pp. 690-695).

Schulz Buschoff, L. M., Akata, E., Bethge, M. & Schulz, E. (2025). Visual cognition in multimodal large language models. *Nature: Machine Intelligence*, 1-12.
<https://doi.org/10.1038/s42256-024-00963-y>

Week 13: Prejudice, Ethics in Machine Learning and Adversarial Examples (04/24)

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Buolamwini, J., & Gebru, T. (2018). Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification. *Proceedings of Machine Learning Research*, 81, 1-15.

Miller, E. J., Steward, B. A., Witkower, Z., Sutherland, C. A. M., Krumhuber, E. V., & Dawel, A. (2023). AI Hyperrealism: Why AI Faces Are Perceived as More Real Than Human Ones. *Psychological Science*, 34(12), 1390-1403.

Zhou, Z., & Firestone, C. (2019). Humans can decipher adversarial images. *Nature Communications*, 10, 1334-1343.

Wallace, E., Rodriguez, P., Feng, S., Yamada, I., & Boyd-Graber J. (2019). Trick Me If You Can: Human-in-the-Loop Generation of Adversarial Examples for Question Answering. *Transactions of the Association for Computational Linguistics*, 7, 387-401.

Costello, T. H., Pennycook, G., & Rand, D. G. (2024). Durably reducing conspiracy beliefs through dialogues with AI. *Science*, 385 (6714), <https://doi.org/10.1126/science.adq1814> .

Optional (not for presentation): Ivanova, A. (2025). Examining cognitive biases in LLMs paper. *Nature: Human Behaviour*, 1-4. <https://doi.org/10.1038/s41562-024-02096-z>

Assignment 4 (Reinforcement Learning) is due on 04/23 at 8:59pm.

Last Discussion Post Due 04/22 at 8:59pm

Week 14: Student project presentations (05/01)

Final papers are due on 05/06 at 8:59pm.